

It is plain common sense that reusing existing resources has the potential to save effort. It was, therefore, important to spend time here describing our lessons learned from attempting to use available knowledge resources and explaining why the current offerings do not directly fulfill the needs of LEIAs. Now we will turn to the resources that LEIAs actually need in order to implement the computational cognitive models underlying them.

3.2 Ontology

A computational ontology is a model of the world designed to foster reasoning by agent systems. The ontology used by LEIAs is organized as a multiple-inheritance hierarchical graph of concepts—OBJECTS and EVENTS—each of which is described using PROPERTYs. Concepts are named using language-independent labels, written in small caps, that resemble English words only for the benefit of English-speaking developers. The actual meaning of a concept is its inventory of property-facet-filler triples.

Facets permit the ontology to include an extra level of detail about property fillers, such as the fact that the most typical colors of a car are white, black, silver, and gray; other normal, but less common, colors are red, blue, brown, and yellow; and rare colors are green and purple. The inventory of facets includes: *default*, which represents the most restricted, highly typical subset of fillers; *sem*, which represents typical selectional restrictions; *relaxable-to*, which represents what is, in principle, possible although not typical; and *value*, which represents not a constraint but an actual, non-overridable value. *Value* is used primarily in episodic memory but applies to a select few properties in the ontology, such as DEFINITION, IS-A, and SUBCLASSES. A sampling of properties from the ontological frame for the event SURGERY illustrates the use of facets.

SURGERY

DEFINITION	value	performing a medical procedure that involves cutting into tissue
IS-A	value	INVASIVE-PROCEDURE
AGENT	default	SURGEON
	sem	MEDICAL-PERSONNEL
	relaxable-to	HUMAN
THEME	default	HUMAN
	sem	ANIMAL

INSTRUMENT	sem	SURGICAL-INSTRUMENT
LOCATION	default	OPERATING-ROOM
	sem	MEDICAL-BUILDING
	relaxable-to	PLACE

The main benefits of writing an ontology in a knowledge representation language rather than a natural language are (a) the absence of ambiguity in the knowledge representation language, which makes the knowledge suitable for automatic reasoning, and (b) the ontology’s reusability across natural languages.

The ontology covers both general and specific domains and currently contains around nine thousand concepts. The bulk of it was compiled some twenty-five years ago, and, in keeping with our lab’s focus on cognitive modeling research rather than application development, only modest enhancements have been made since. We use the ontology as a research tool and enhance it to test the ever-growing inventory of microtheories—including the hypothesis that LEIAs can acquire ontology independently through dialog, experience, and reading.

The example of SURGERY illustrates the simplest kind of ontological structure. The expressive power of the ontology is actually far greater, and its content far richer, since the ontology is intended to support multiple agent functionalities including simulation, reasoning, learning, teaching, explaining, and beyond.

3.2.1 Properties

Every OBJECT and EVENT is described using PROPERTYS. The inventory of PROPERTYS essentially supplies the axiomatic layer of the ontology’s representational system. PROPERTYS are characterized by their DOMAIN (constraints on the sets of concepts for which they are defined) and their RANGE (their value sets). The LEIA ontology includes several types of PROPERTYS:

- IS-A and SUBCLASSES indicate the concept’s placement in the inheritance hierarchy. Multiple inheritance is permitted but not overused, and rarely does a concept have more than two parents.
- RELATIONS indicate relationships among OBJECTS and EVENTS. The DOMAIN and RANGE of RELATIONS are, therefore, filled by OBJECTS and/or EVENTS. Examples include the case roles (e.g, AGENT, THEME⁷), spatial relations (e.g., ABOVE, NEXT-TO), HAS-OBJECT-AS-PART, CAUSED-BY, and so on. All RELATIONS have

- inverses: for example, the inverse of AGENT is AGENT-OF, the inverse of ABOVE-AND-TOUCHING is BELOW-AND-TOUCHING, and the inverse of CAUSED-BY is EFFECT.
- SCALAR-ATTRIBUTES are properties of OBJECTS and EVENTS that can be expressed by numbers or ranges of numbers, such as COST, VELOCITY, and WEIGHT. Values can be actual—for example, *180 pounds* is “WEIGHT (180 MEASURED-IN POUND)”—or they can be expressed on the abstract scale {0,1}—for example, *heavy* is “WEIGHT .8” and *light* is “WEIGHT .2.”
 - LITERAL-ATTRIBUTES are properties of OBJECTS and EVENTS whose fillers are represented by uninterpreted literals. For example, the property MARITAL-STATUS has the literal fillers *single*, *married*, *divorced*, and *widowed*.
 - SUBEVENTS holds ontological scripts, also known as complex events (section 3.2.4).
 - SEMANTIC-EXPANSION holds concept-based descriptions of complex properties (section 3.2.1).
 - The DEFINITION field holds a natural language string that explains the concept. Definitions are used by developers as well as by LEIAs for purposes of explanation.

Ontological PROPERTYs can also function as ABSTRACT-OBJECTs: *Friendliness is important; Color livens up a house*. Property-based abstract nouns are recorded in the lexicon as an ABSTRACT-OBJECT with a RELATION to the meaning of the adjective. Since the adjective *friendly* is described as “FRIENDLINESS .8,” the noun *friendliness* is described as:

ABSTRACT-OBJECT	
RELATION	FRIENDLINESS-1
FRIENDLINESS-1	
RANGE	.8

Treating nominal uses of properties as ABSTRACT-OBJECTs allows them to participate in larger propositions in the normal way—they can be modified, evaluated as case-role fillers, and so on. For example, the meaning representation for **Friendliness is important** is:

ABSTRACT-OBJECT-1	
RELATION	FRIENDLINESS-1
IMPORTANCE	.8
FRIENDLINESS-1	
RANGE	.8